IoT-Based HVAC Monitoring System for Smart Factory

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***Abstract*—This paper presents an Internet of Things (IoT) Based HVAC monitoring platform for smart factory. A system model of energy-efficient Heating, Ventilation and Air Conditioning (HVAC) management in smart factory, which is useful for monitoring, controlling and saving energy. The hardware implementation was done using prototype built with Raspberry pi model B, temperature sensor, humidity sensor, current sensor and Micro-controller to gather and monitor data related to the HVAC for a typical smart factory. The system can collect local/raw data needed for the analysis using an off-the-shelf Big data analytics software. MATLAB pattern recognition toolbox was used to test, train and vaildate the behaviour of data collected and processed using R programming software. The paper presents a promising insights into the combined usage of IoT technologies, Big data analytics and neural network to solve future generation smart factory problems of HVAC management**

***Index Terms*—Big data, HVAC, IoT, pattern recognition, raspberry**

I. Introduction

Heating, Ventilation and Air-Conditioning (HVAC) has been shown to account for 30-40 percentage of most total electricity bills and the need to lower cost is imperative both for smart factories and commercial buildings. Another need for intelligent HVAC is to ensure optimization of these resources despite the need to cut cost. There is need for example for room temperature optimization such that rooms are kept at the needed temperature at any point in time. To fix the dual challenge of HVAC cost [1] and optimization, an intelligent monitoring system is proposed. Features include-sensing the parameters indoor (Temperature, humidity and carbon dioxide), detects human presence, has efficient algorithm and provides feedback via web application. Benefits include Feedback, control and optimization of HVAC , reduced operational cost.

Internet of Things (IoT) portrays a rising pattern where an enormous number of embedded gadgets also described as things that are associated with the Internet to such an extent that they speak with individuals and different things. They are readily applicable to various sectors including for manufacturing, smart farm and industrial scenario [2]. IoT technologies allows for sensing, monitoring and control of HVAC in households on a large scale.

With the proliferation of IoT and growth in smart factory applications [3], it has been shown that the numerous advantages of smart home(or factory) or home automation systems (HAS) comes with a cost. In [4], Authors demonstrated that all things considered, HAS may expend more than 33% of the yearly energy utilized in a medium sized home. In addition to the problem of cost of energy consumption, rising data collection also poses a challenge to storage, organization, analysis and visualization of IoT data. IoT and big data solutions comes readily handy as a solution to this problem. This will allow the smart factory users to have needed feedback and control over HVAC situations in their home just as it will assist energy service providers flexible and on-demand supply [5].

This paper illustrates a prototype of IoT based monitoring system of HVAC as well as providing analytic utilizing R programming for Big Data analysis. The remainder of the paper is organized as: following this Section I is Section II, where a brief review of similar works were discussed with a view to show trends on the subject matter. In Section III, a block diagram description of the proposed systems requirements. This is followed by Section IV where the prototype is presented. Results and discussion of the findings are presented in Section V while paper is concluded in Section VI.

II. Related Works

The use of IoT based monitoring system has attracted research interest. In [6], a brokerless architecture was adopted to provide real time monitoring of water quality taking note of the temperature, water pH and dissolved oxygen. The authors however limited their works to data capturing without consideration for a large scale scenario where the need for big data analytics will be imperative. The pervasive and ubiquitous usage of IoT with large volume of data has made the incorporation of Big data very important in recent time [5].

In order to gain useful insight into the relationship between indoor temperature and ventilation patterns in building, [7] adopted data mining techniques with the purpose of monitoring and controlling smart building ventilation subsystem. The raw data can be captured by IoT sensors and data aggregated (compressed) prior to classification using machine learning. The results becomes very useful for understanding the behaviour of equipment in the building and makes fault finding and maintenance very easy. This explains the rising academic and industry interest in monitoring and evaluating

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HVAC conditioning strategy of buildings needed to avoid waste.

IoT as an emerging technology for analysing and inference making on indoor air quality as it affects indoor environmental and health condition was studied in [8]. The authors used IoT sensors to capture the air quality and using Pearson correlation coefficient, drew a correlation between air quality and level of health hazards occupants of a room are exposed to. However, their works were limited as it can not handle big data scenario. It is in this light, that the work of [9] becomes relevant as they did an extensive review of major IoT products and technology that will shape future generation monitoring systems.

One key feature listed was the need to automate the process of automation of data collection and analysis using cloud and big data computing. In a recent work [10], machine learning was adopted to learn driver's usage pattern with a view to predict the HVAC value for a vehicle serves as a feedback and control for the maintenance of HVAC features of the vehicle. The difference however, the paper target system was vehicles and not smart factory as proposed in this paper.

III. System Model

In the HVAC monitoring system, we need to activate the sensors first. Humidity, temperature, and current sensor must be connected to Raspberry Pi 3. Here Raspberry Pi 3 is used as sink node and MQTT protocol is applied for transmitting collected data from sensors to smartphone. MQTT has low power consumption and light weighted as it is a binary protocol. The Wifi module is used as a network connectivity for the transmission of all information from sensors and passing to cloud service for storing data.

Here we have used ThingSpeak as Application Interface(API), a software component to collect and store sensor information which is further visualized and analyzed by R programming for big data analytics. We also have used laptop or PC for controlling and updating programs for Raspberry Pi as well as monitoring the energy consumption of the sensors.

Humidity and temperature sensor collect weather conditions of home and transfer information using MQTT protocol. A notification of current state such as air conditioning and energy consumption of sensors in home can be shown in smartphone. And after monitoring this state, for example, if the room or factory floor is too hot or too cold, user can easily control system using relay switch. That's how records of the power consumption status and control over the sensors through cloud system can be achieved. The system model is summarized in Fig. 1.

IV. Experiment and Design Set up

To make the implementation of IoT based systems cheaper, several platforms such as Ardruino, Galileo board, Zigbee, Xbee, Raspberry Pi and Intel Edison board are now in use [11]. The authors in [11] designed a system that counts the number of people that enters and exits a room and used IoT based system to design how best to manage the HVAC since room temperatures rises as the number of people increases in a room.

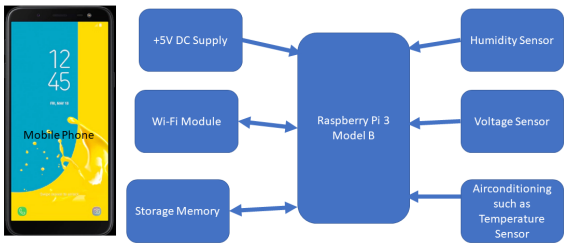

Fig. 1. A block diagram of the set up of the IoT-Based HVAC Monitoring System showing the flow of data from the HVAC sensors and interaction with the Raspberry p. The smart factory user gains insight, feedback and control using the mobile phone

However, their proposal was just limited to one of the HVAC features which is heating (H). Fig. 2 displays the schematic diagram of the experiment set up.

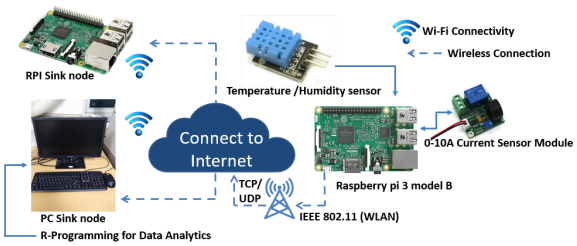

Fig. 2. Experiment set up of the IoT-Based HVAC Monitoring System showing typical sensors, Raspberry pi and the off the shelf Big data analytic software for smart factory

A. Description of components

In this section, we give a brief description of the components as made up in Fig. IV.

1) Sink Node: Intel® Core ™ i7 processor with a 24GB RAM and 64-bit Windows 7 operating System, clocked at 3.40Ghz, and one Rasp-berry pi 3 model B were utilized as sink nodes in the implementation. It is called as base station, involved collecting, processing, and analyzing the gathered sensed data from different sensors. To save energy consumption in the whole network, the best location of sink node should be properly chosen [12].

2) Source Node: The source node is considered to monitor sensed information around network. Source node detects activity in smart factory and sends information intermittently to the sink node based on multi hop routing communication scheme [13]. We have employed Raspberry pi 3 model B (Rpi-B) was used in this set up to serve this purpose.

3) Raspberry Pi 3 Model B: It is a credit-card sized computer with a 1.2 GHz 64-bit quad core processor launched in February 2016. This device connects to a PC Screen, PC or TV employing Bluetooth, on-board 802.11n Wi-Fi, and USB boot capabilities [14].

4) Humidity and Temperature Sensor: We have used DHT11 among these effective sensors such as SHT21, DHT22, S17021, HDC1080 and so on. It is a budget friendly in

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comparison with DHT22 and operates sensing moisture/ temperature for weather conditioning. Sensor consists of a thermistor and capacitive humidity sensing element. Changing capacitance value creates alteration in humidity levels and a basic IC converts analog value to digital form.

5) Current Sensor: : ACS712 is used for monitoring HVAC system determining and calculating current with generating a signal relative to distinguished current either in the form of digital output or simple voltage from the hardware setup. This sensor located at IC based on copper conduction path, senses produced magnetic field while current flowing and generates a voltage signal.

V. Performance Evaluation

In order to show the typical behaviour of our experiment set up, this paper adopted dataset from [15]. The dataset is available on IEEE dataport and useful for statistical modeling of thermal or heating management of buildings. Data processing was done using a combination of R and MATLAB. The R programming language was used for data analytics, visualization and reporting in the absence of online data analytic application such as ThingSpeak.

HVAC of the smart factory or indeed any building comprises of data regarding the heating and variability of the systems [16]. Heating consumption has been shown to be a function of several factors such as wall and roof material, floor numbers, type of heating system (open or closed). About 700,036 data was collected within the period under review and sorted into various dependent and independent factors. Using R programming, features needed for this paper were extracted such as HVAC type, floor area, type of material used in building the factory, and temperature.

For the classification of material type, all smart factory walls built with panel were set at 1 while those built with bricks were set at 2, wood materials were set at 3, block set at 4 and monolith set at 5 respectively. The modified dataset is then trained and analysed using MATLAB. Fig. 3 shows the classification of the type of material used for the smart factory wall

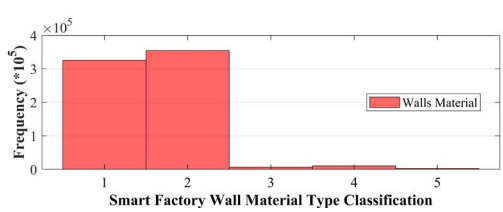

Fig. 3. Smart Factory building wall material type classification. Recall that the type of material used for the wall affects the HVAC reading. Here, the least was monolith while most material walls were made of bricks as shown by frequency of 2

In the same vein, Fig. 4 shows the range of the temperature of the building taken for the 700,036 times. It can be observed that temperature ranged between -30°and 20°. Fig. 5 depicts the summary of the smart factory area of data collected.

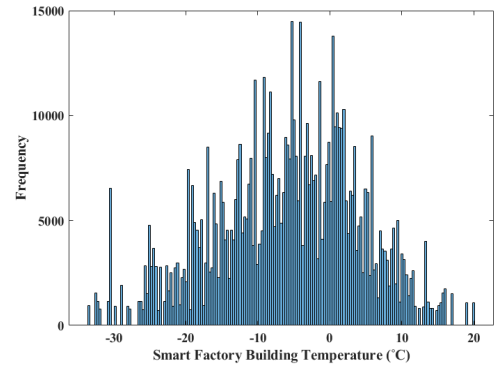

Fig. 4. Smart Factory temperature collected and analysed using MATLAB. About 700,036 data collected. It can be seen that the highest temperature recorded was between -10°and 0°

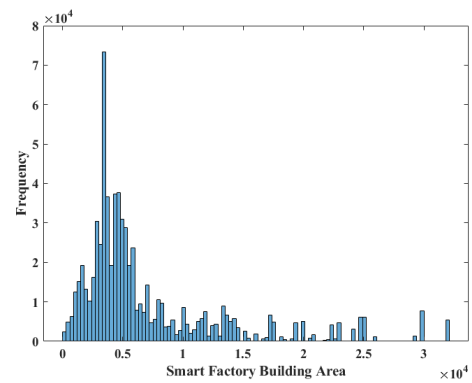

Fig. 5. Smart Factory Building area collected and analysed using MATLAB. About 700,036 data collected. It can be seen that the highest area is 4401

A. Machine Learning Evaluation

Using Machine learning toolbox in MATLAB, we trained the data and used in carrying out pattern recognition (See Fig. 6).

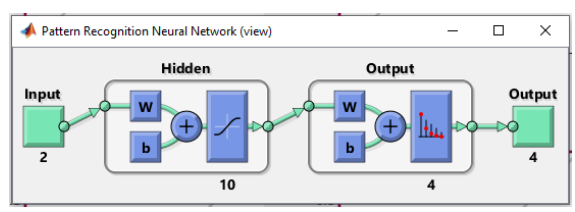

Fig. 6. The Neural Network Model Showing the inputs, hidden layer and output Scenario

To optimize the neural network training, a randomly divided sample of up to 1000 was used. These were further divided into 700 samples for training, 150 for validation and the rest 150 percentage for testing. Number of hidden neurons was placed at 10. Training was done using conjugate gradient back-propagation. The number of Epoch was 30 iterations. Results are summarized in Fig. 7 and Fig. 8 respectively for want of space.

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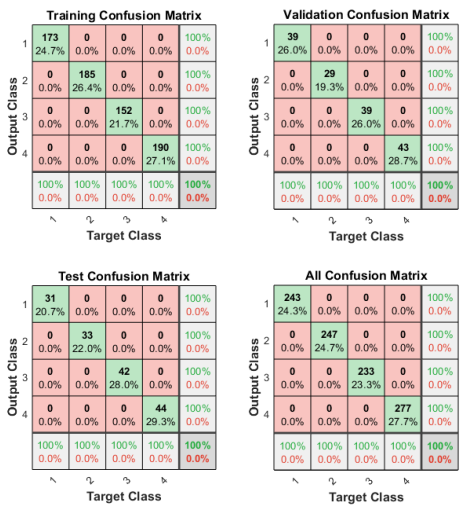

Fig. 7. Confusion Matrix of the pattern recognition using MATLAB Pattern Recognition ToolBox.

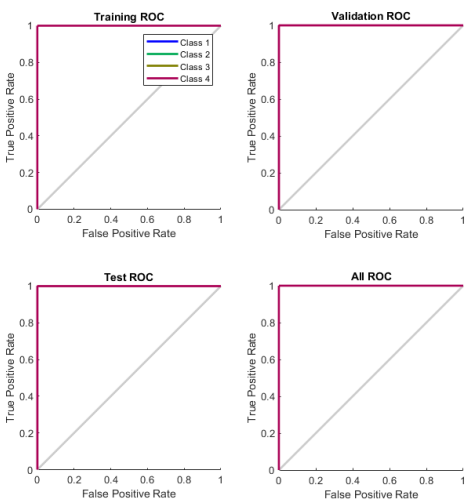

Fig. 8. Training, Validation and testing ROC Result using MATLAB Pattern Recognition ToolBox.

VI. Conclusion

The paper demonstrated an IoT-Based HVAC monitoring system for capturing humidity, ventilation and air-condition in a smart factory. The current sensor allows for the monitoring of energy consumption in the smart factory. The results shows the potential of the set up as a support system for low latency and cheaper implementation of energy metering system in the era of explosion of IoT devices in smart factory. Future works will involve a larger test-bed and factoring more complex scenario typical of an ideal smart factory or smart home.

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